

1. E-Commerce Billing Engine

Create a billing engine for an online store using basic OOP and method

Requirements:
concepts.

1. Create a class Item with:
 - id
 - name
 - unitPrice
2. Create a class Bill with:
 - Array of purchased Item IDs
 - Quantity array
 - Method calculateTotal()
3. The program should:
 - Accept user input for items purchased

Item	Quantity	Price per unit
Pen	2	₹ 20
Book	1	₹ 50

Solution :

In this example, the customer purchases 2 Pens priced at ₹20 each and 1 Book priced at ₹50. The program calculates and displays the final bill amount based on the purchased quantities and unit prices.

Java Coding:

```
import java.util.*; class Item {  
    int id;  
    String name;  
    double unitPrice;  
    Item(int id, String name, double unitPrice) {  
        this.id = id;  
        this.name = name; this.unitPrice = unitPrice;  
    }  
}
```

```

class Bill {
int[] itemIds;
int[] quantities;
Item[] items;
Bill(int[] itemIds, int[] quantities, Item[] items) {
this.itemIds = itemIds; this.quantities = quantities; this.items = items;
}
double calculateTotal() {
double total = 0;
for (int i = 0; i < itemIds.length; i++) {
for (Item item : items) {
if (item.id == itemIds[i]) {

total += item.unitPrice * quantities[i]; }
} }
return total; }
}

public class EcommerceBillingEngine {
public static void main(String[] args) { // Available Items
Item[] items = {
new Item(1, "Pen", 20),
new Item(2, "Book", 50) };
// Purchased Items
int[] itemIds = {1, 2};
int[] quantities = {2, 1};
Bill bill = new Bill(itemIds, quantities, items); System.out.println("----- BILL -----");
System.out.println("Pen x 2 = Rs.40"); System.out.println("Book x 1 = Rs.50");
System.out.println("-----");
System.out.println("Total Amount = Rs." + bill.calculateTotal());
} }

```

Output:

```

----- BILL -----
Pen x2=Rs.40
Book x 1 = Rs.50 -----
Total Amount = Rs.90.0

```